

PROJECT #1 KEY

Step 1: Write a function for a 1D position as a function of time + initial conditions (for constant acceleration).

Start w/ known relationships like $a = \frac{dv}{dt}$

Integrate both sides.

acceleration
is const.

$$\int a \, dt = \int dv$$

$$at = v(t) - v_0$$

$$v(t) = v_0 + at$$

Recall:

Fundamental Theorem
of Calculus:

$$\int_a^b f \, df = f(b) - f(a)$$

for some function f .

Rearrange.

Recall the relationship $v = \frac{dx}{dt}$.

Substitute $v = \frac{dx}{dt}$.

$$\frac{dx}{dt} = v_0 + at$$

Integrate again.

$$\int \frac{dx}{dt} = \int (v_0 + at)$$

$$\int dx = \int (v_0 + at) \, dt$$

$$x(t) - x_0 = v_0 t + \frac{1}{2} at^2$$

$$x(t) = x_0 + v_0 t + \frac{1}{2} at^2$$

To write as a function in Python:

1. Identify input variables: x_0, v_0, t, a
2. Identify desired output: $x(t)$
3. Identify equation(s) needed: $x(t) = x_0 + v_0 t + \frac{1}{2} at^2$
4. Fill in function outline with correct variables.

```
def function_name(input_variable):  
    # Description of function operations.  
    return output_variable
```

→

```
def position(x0, v0, t, a):  
    # Returns 1D position at a given time  
    return x0 + v0 * t + (1/2) * a * t ** 2
```

Step 2: Write a function for a 1D velocity as a function of time and initial conditions (for constant acceleration).

Similar methods to Step 1.

Start with known relationship: $a = \frac{dv}{dt}$

Separate & Integrate

$$\int a \, dt = \int dv$$

$$at = v(t) - v_0$$

Rearrange.

$$v(t) = v_0 + at$$

Write as python function:

```
def velocity(v0, a, t):
```

```
    *Returns 1D velocity for a given initial velocity, acceleration, and time.  
    return v0 + a * t
```

Step 3: Fixing $a_y = 9.8 \text{ m/s}^2$ and $a_x = 0$, write code for plotting the variables x, y, v_x, v_y .

Recall velocity equations: $v_x(t) = v_{0x} t + \frac{1}{2} a_x t^2$ also $x'(t)$

$$v_y(t) = v_{0y} t + \frac{1}{2} a_y t^2$$

$v_{0x} = v_{0y} = v_0 = 0$ because the initial velocity, $v_0 = 0$

$$v_x(t) = \cancel{\frac{1}{2} a_x t^2} \quad a_x = 0$$

$$v_y(t) = \frac{1}{2} a_y t^2 \quad a_y = 9.8 \text{ m/s}^2$$

Plot v_x and v_y versus time.

↳ See google colab key.